E-GOVERNANCE WEB INFRASTRUCTURE OBSERVATORY

BharatGov Access

A Census-Scale In-Country Web Observatory and Empirical Benchmark of India's Public Digital Infrastructure

CENSUS REGISTRY

12,092

Central, State, District & ULB Portals

IN-COUNTRY INGESTION

10,247

Active Evaluated Portals (84.7%)

STANDARDIZED SCHEMA

5 Pillars

Exactly 10 Focused Sub-Features

CLOUD INFRASTRUCTURE

Bangalore

6 vCPU / 60 GB RAM / Zero Geo-Block

RESEARCH TEAM & ACADEMIC PROJECT DETAILS

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Live Observatory Dashboard: https://quality-ampland-ashley-ought.trycloudflare.com

ACADEMIC BACKGROUND

Empirical Motivation & Academic Research Problem

1. The Scale of Digital India

Public Portals as Critical Infrastructure

- Over **10,000 public portals** serve 900+ million citizens across 36 States/UTs.
- Portals deliver Direct Benefit Transfer (DBT), land registries, public exams, and grievance redressal.
- Digital exclusion occurs when public websites fail accessibility, language, or performance standards.

2. The Research Gap

Limitations of Existing Audits

- **Small Arbitrary Samples:** Existing studies inspect <100 homepages, ignoring district administrations.
- **Foreign Geo-Blocking:** Overseas crawlers face state perimeter drops (66%+ false 502/504 errors).
- **Feature Bloat:** 70+ noisy columns fail to isolate core indicators or decouple server reachability.

3. Academic Contribution

Census-Scale Domestic Observatory

- **In-Country Bangalore Node:** Zero perimeter blocking, authentic citizen-experienced latencies.
- **Clean 5-Pillar / 10-Sub-Feature Model:** Aligned with W3C (WCAG 2.1 AA) and MeitY (GIGW 3.0).
- **Publication Target:** Empirical conference/journal paper on India's digital public infrastructure.

Research Question: How conformant is India's multi-tiered public web infrastructure when audited domestically across WCAG, CWV, and CERT-In standards?

EVALUATION FRAMEWORK

Standardized Architecture: 5 Main Features & 10 Sub-Features

Pillar 1: Accessibility

WCAG 2.1 AA / GIGW 3.0

`sub_accessibility_score`

0–100 score: Image Alt text, form labels, HTML lang tags.

`sub_has_gigw_toolbar`

Boolean: Font resizers A+/A-, contrast switcher, skip-to-content.

Pillar 2: Performance

Core Web Vitals & Latency

`sub_response_latency_ms`

TTFB / Network round-trip latency in ms from Bangalore.

`sub_lcp_est_sec`

Estimated Largest Contentful Paint in seconds.

Pillar 3: Security

CERT-In Cyber Hardening

`sub_https_enforced`

Boolean: Strict TLS/HTTPS encryption enforcement.

`sub_security_headers_score`

0–100 score: HSTS, CSP, X-Frame, X-Content-Type.

Pillar 4: Inclusivity

Digital India Reach

`sub_is_multilingual`

Boolean: Content in 22 Scheduled Indian regional scripts.

`sub_mobile_responsive`

Boolean: Viewport meta tag width=device-width.

Pillar 5: Architecture

Citizen Interactivity

`sub_cms_architecture`

Categorical: S3WaaS Drupal, Custom CMS, ASP.NET, SPA.

`sub_has_citizen_services`

Boolean: Native Search and CPGRAMS Grievance links.

Composite Index: `observatory_maturity_score` (0–100 Weighted Score) synthesizes all 5 pillars into a single defensible benchmark indicator.

20 Standard Columns

ENGINEERING & INFRASTRUCTURE

Measurement Methodology & In-Country Cloud Pipeline

1. Domestic Cloud Instance

Bangalore Datacenter Node

- Dedicated Ubuntu 22.04 LTS cloud VM in Bangalore, India.
- Hardware: 6 vCPUs, 60 GB RAM, 350 GB NVMe storage.
- **Eliminates Perimeter Blocks:** Direct egress within Indian ISP/RailTel peering fabric resolves false-positive 502/504 blocks.

2. High-Throughput Engine

Async Concurrency Pipeline

- 64 concurrent async workers utilizing httpx & BeautifulSoup.
- HTTP/1.1 & HTTP/2 automated protocol negotiation.
- **Throughput:** Crawled 2,390 empirical baseline domains in 182.8 seconds (13.1 domains/second).

3. Decoupled Validation

Strict Measurement Rigor

- Registry validation is strictly decoupled from server reachability.
- 10,247 portals active and measured; 1,845 verified inactive/offline with explicit diagnostic reasons.
- **Zero Misleading Numbers:** Failed requests are stored as null/NaN, never as false accessibility zeros.

Verification: Bangalore cloud measurements completely eliminated perimeter drops, reducing latencies to authentic 1.18s domestic TTFB.

EMPIRICAL RESULTS

Empirical Findings Across 12,092 Portals

Accessibility & S3WaaS Impact

Framework Standardization Triumph

- **93.8 / 100** National Average Accessibility score across active portals.
- **63.2%** of active portals (6,480) run on NIC's S3WaaS (Drupal) framework.
- S3WaaS portals achieve **96.1 / 100** accessibility and 98% GIGW toolbar adoption.
- Legacy ASP.NET portals lag significantly at **81.5 / 100**.

Security & Cryptography Gap

CERT-In Compliance Deficit

- **100.0%** Portals enforce TLS/HTTPS encryption baseline.
- Average Security Header Score is only **68.0 / 100**.
- Only **44.5%** enforce HSTS (Strict-Transport-Security).
- Only **35.2%** declare Content-Security-Policy (CSP).
- High exposure to SSL-stripping on legacy state gateways.

State Web Maturity Rankings

Top Performing States (0–100)

- **West Bengal:** 81.2 (96.4 Acc, 98% GIGW)
- **Nagaland:** 80.5 (97.1 Acc, 100% GIGW)
- **Haryana:** 79.8 (96.8 Acc, 97% GIGW)
- **Mizoram:** 79.5 (95.8 Acc, 92% GIGW)
- **Uttar Pradesh:** 78.4 (Comprehensive district rollout)
- **Central Government:** 78.2 (highest security: 85.2)

Policy Implication: Migrating remaining state portals to S3WaaS immediately resolves accessibility gaps; enforcing HSTS/CSP is the critical next milestone.

National Benchmark

PRODUCT & RESEARCH ROADMAP

Live Observatory Product & Academic Research Roadmap

1. Live Public Observatory

Production Dashboard

- Deployed via Cloudflare Tunnel: **quality-ampland-ashley-ought.trycloudflare.com**
- Interactive 5-pillar radar comparisons across Central, State, and District tiers.
- Policy scorecards for all 36 States/UTs.
- Searchable Registry of all 12,092 government portals with instant CSV export.

2. Open Data Artifacts

FAIR Open Science Releases

- Standardized 20-column schema (Parquet, CSV, JSON).
- Released under Creative Commons (CC-BY-4.0).
- Open Data Deposit prepared for Kaggle & Hugging Face.
- Automated Python analysis quickstart included.

3. Research Paper Publication

Targeting Tier-1 Venues

- **Paper Title:** "BharatGov Access: A Census-Scale In-Country Web Observatory and Empirical Benchmark of India's Public Digital Infrastructure".
- **Contributions:** Census-scale DPI analysis, empirical proof of S3WaaS framework impact, and policy blueprint.
- Target: ACM COMPASS, IEEE Internet Computing, or GIQ.

Checkpoint 02 Complete: Census 5-Pillar / 10-Sub-Feature Dataset · In-Country Bangalore Ingestion · Live Public Observatory Dashboard